

Step 1: Create the Customers Table

None

```
CREATE TABLE Customers (  
    customer_id INT PRIMARY KEY,  
    customer_name VARCHAR(50),  
    city VARCHAR(50)  
);
```

```
INSERT INTO Customers  
VALUES  
(1, 'John', 'Chennai'),  
(2, 'Alice', 'Bangalore'),  
(3, 'David', 'Hyderabad'),  
(4, 'Sophia', 'Mumbai'),  
(5, 'Emma', 'Delhi');
```

Step 2: Create the Orders Table

None

```
CREATE TABLE Orders (  
    order_id INT PRIMARY KEY,  
    customer_id INT,  
    product VARCHAR(50),  
    amount DECIMAL(10,2)  
);
```

```
INSERT INTO Orders  
VALUES  
(101, 1, 'Laptop', 65000),  
(102, 2, 'Mobile', 25000),  
(103, 1, 'Mouse', 800),  
(104, 3, 'Keyboard', 1500),  
(105, 6, 'Monitor', 12000);
```

Activity 1

Question: Display the customer name and the product they ordered.

Hint:

- Use **INNER JOIN**.
 - Join the **Customers** and **Orders** tables using **customer_id**.
 - Display **customer_name** and **product**.
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Activity 2

Question: Display all customers along with their orders.

Hint:

- Use **LEFT JOIN**.
 - The **Customers** table should be the left table.
 - Customers without orders should also appear.
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Activity 3

Question: Display all orders, even if the customer details are not available.

Hint:

- Use **RIGHT JOIN**.
 - The **Orders** table should be the right table.
 - Orders without matching customers should also appear.
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Activity 4

Question: Find customers who have not placed any orders.

Hint:

- Use **LEFT JOIN**.
 - Find rows where the order information is **NULL**.
 - Check the **order_id** column for **NULL**.
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Activity 5

Question: Find orders that do not have a matching customer.

Hint:

- Use **RIGHT JOIN**.
 - Find rows where the customer information is **NULL**.
 - Check the **customer_id** from the **Customers** table.
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Activity 6

Question: Display the customer name, product, and order amount.

Hint:

- Use **INNER JOIN**.
 - Select columns from both tables.
 - Join using **customer_id**.
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Activity 7

Question: Display customers who purchased a Laptop.

Hint:

- Use **INNER JOIN**.
 - Filter the result using the **WHERE** clause.
 - Check the **product** column.
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Activity 8

Question: Calculate the total purchase amount for each customer.

Hint:

- Use **LEFT JOIN**.
 - Use the **SUM()** aggregation function.
 - Group the results by customer name.
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Activity 9

Question: Display customers whose total purchase amount is greater than ₹30,000.

Hint:

- Use **INNER JOIN**.
 - Use **SUM()** and **GROUP BY**.
 - Filter the aggregated result using the **HAVING** clause.
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Activity 10

Question: Display every possible combination of customers and products.

Hint:

- Use **CROSS JOIN**.
- No **ON** condition is required.

- Every customer should be paired with every product.
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Bonus Activity 1

Question: Count the number of orders placed by each customer.

Hint:

- Use **LEFT JOIN**.
 - Use the **COUNT()** function.
 - Group the result by customer name.
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Bonus Activity 2

Question: Display customers from Chennai who have placed orders.

Hint:

- Use **INNER JOIN**.
 - Filter using the **WHERE** clause.
 - Check the **city** column.
-

Bonus Activity 3

Question: Find the highest order amount for each customer.

Hint:

- Use **INNER JOIN**.
- Use the **MAX()** function.
- Group the result by customer name.

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